Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 9:05 PM

Agenda

* Variance , Standard Deviation
* Gaussian Distribution (or) Normal Distribution
  - ☆ 68 | 95 | 99.7 Rule (Emperical Rule)
  - ☆ Z - Score
  - ☆ Computing Probability

** Assignments
- Problem

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 9:08 PM

Mean
Median
Mode

Range ⇒ max − min

Selway Drawid

IQR ⇒ $Q_3 - Q_1$

10000

Height - weight

25 %
50 %
75 %
⇒

2 / 2

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 9:25 PM

CDF

fraction shorter than

0.75

0.5

0.25

y

25 %.

75 %.

50 %.

54 54.1 63. 66.3 69. 79 80.2 80.3 90

60.2

10,000

x-axis

Height inches

3 / 4

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 7 Jul 9:21 PM

CDF

$F(x) = y$

10000

25%. 63.3

CDF

63.3

0.25

95%. percentile → 63.3

10 K

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 9:37 PM

⇒ you have to guess his height ?

① 68 inches
64 inches
64
] error → 4 URS

② 63 inches
68 inches
] Error → 5 5RS

③ 55 (ठा) 25

5 / 5

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 9:45 PM

One way to measure Error

(Actual value − guess)^2 ⇒ Mean Squared Error

Mean

MSE

10

$(H_1 - \mu)^2 + (H_2 - \mu)^2 + (H_3 - \mu)^2 - \cdots$

$(H_{10} - \mu^2)$

Variance =

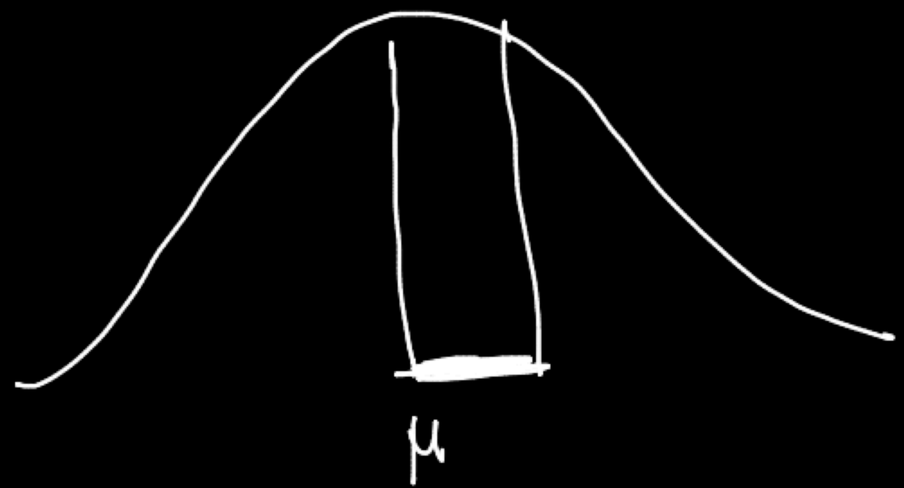
$$\text{n datasets Variance} = \frac{\sum_{i=1}^{n} (H_i - \mu)^2}{n}$$

10

6 / 6

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 9:51 PM

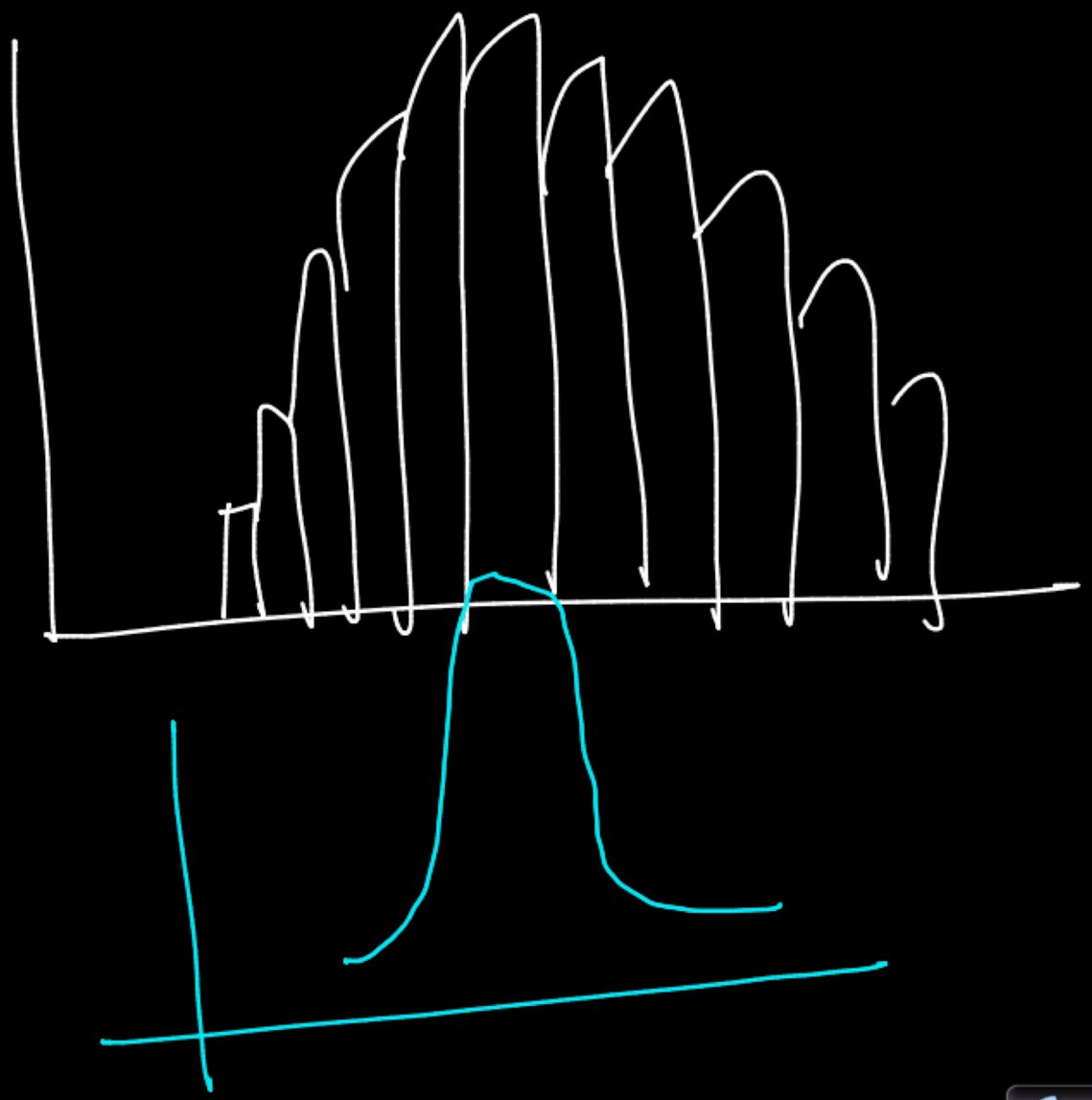
$$\text{variance} = \sigma^2 = \frac{\sum_{i=1}^{n} (H_i - \mu)^2}{n}$$



$$\text{std deviation} = \sqrt{\text{variance}}$$

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 7 Jul 9:53 PM

Gaussian distribution

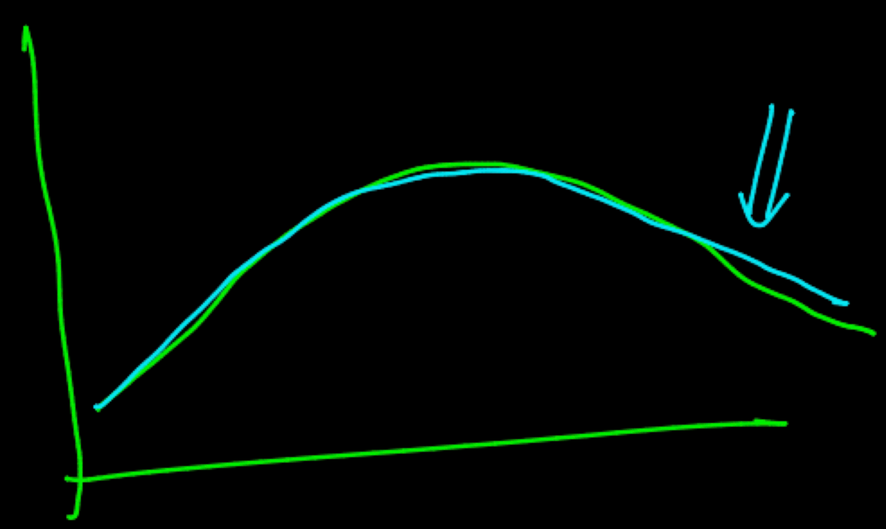


$\mu = 65$ inches



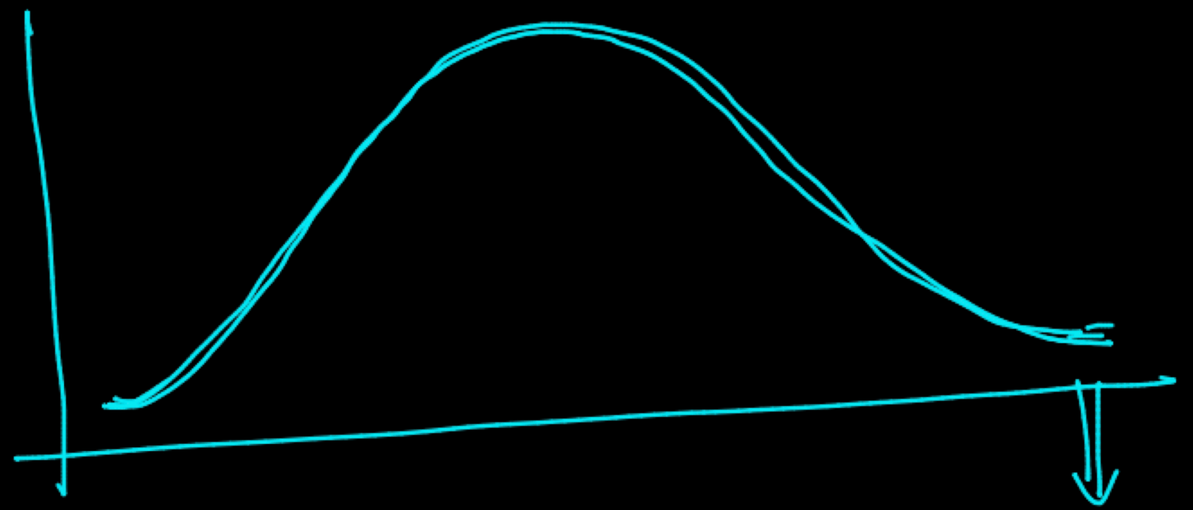
$= 2.5$ inches

spread



Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 7 Jul 9:55 PM

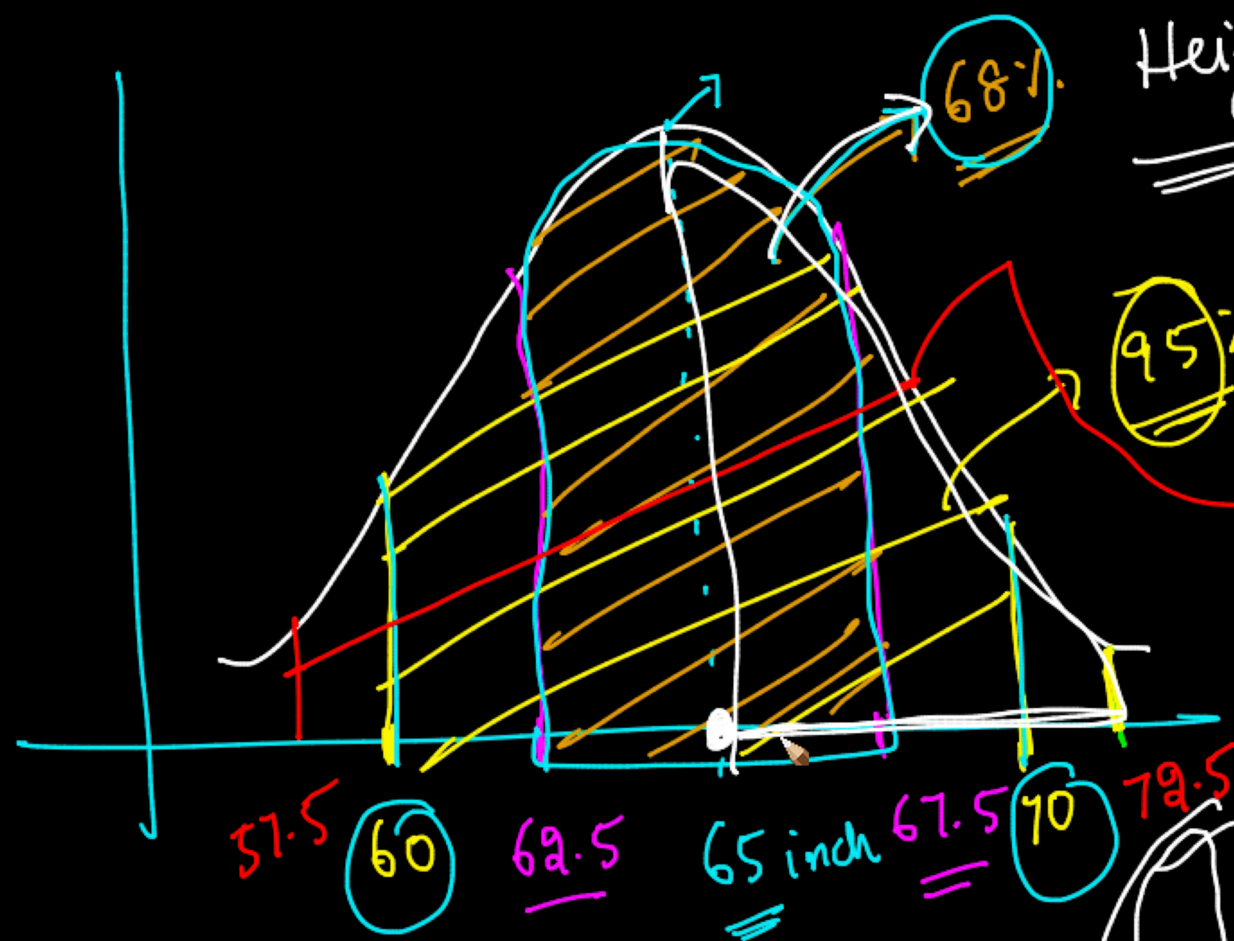
Hv
w
Sales
etc



central limit theorem

9 / 9

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 10:05 PM



Heights

$\mu = 65$

$\sigma = 2.5$

$\Rightarrow$ 1 std dev away from the mean

Empirical 1 σ $\rightarrow$ 68 - 95 - 99.7 Rule

Derive

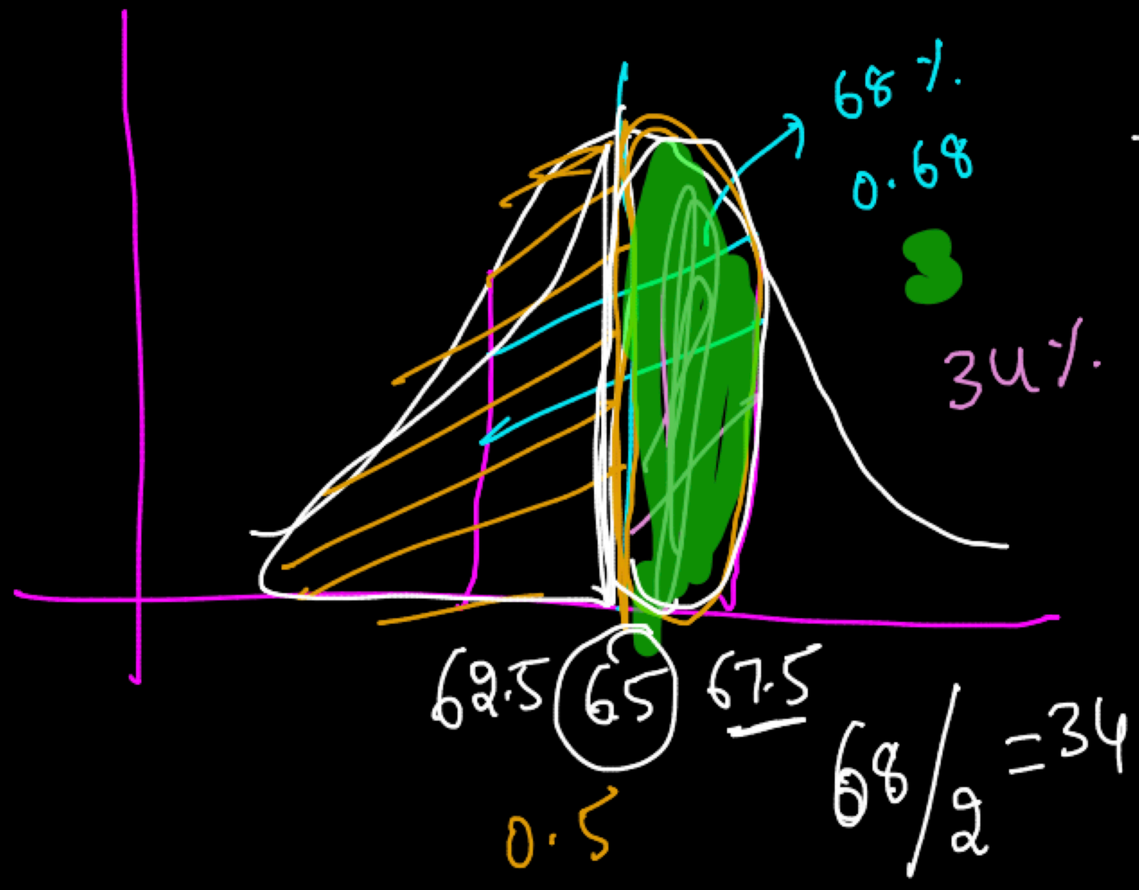


Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 10:12 PM

$\mu = 65$ inches

$\sigma = 2.5$ inches

people are shorter than 67.5 ?



=> Fraction of people shorter than given 65 inches.

0.5

=> Fraction of people

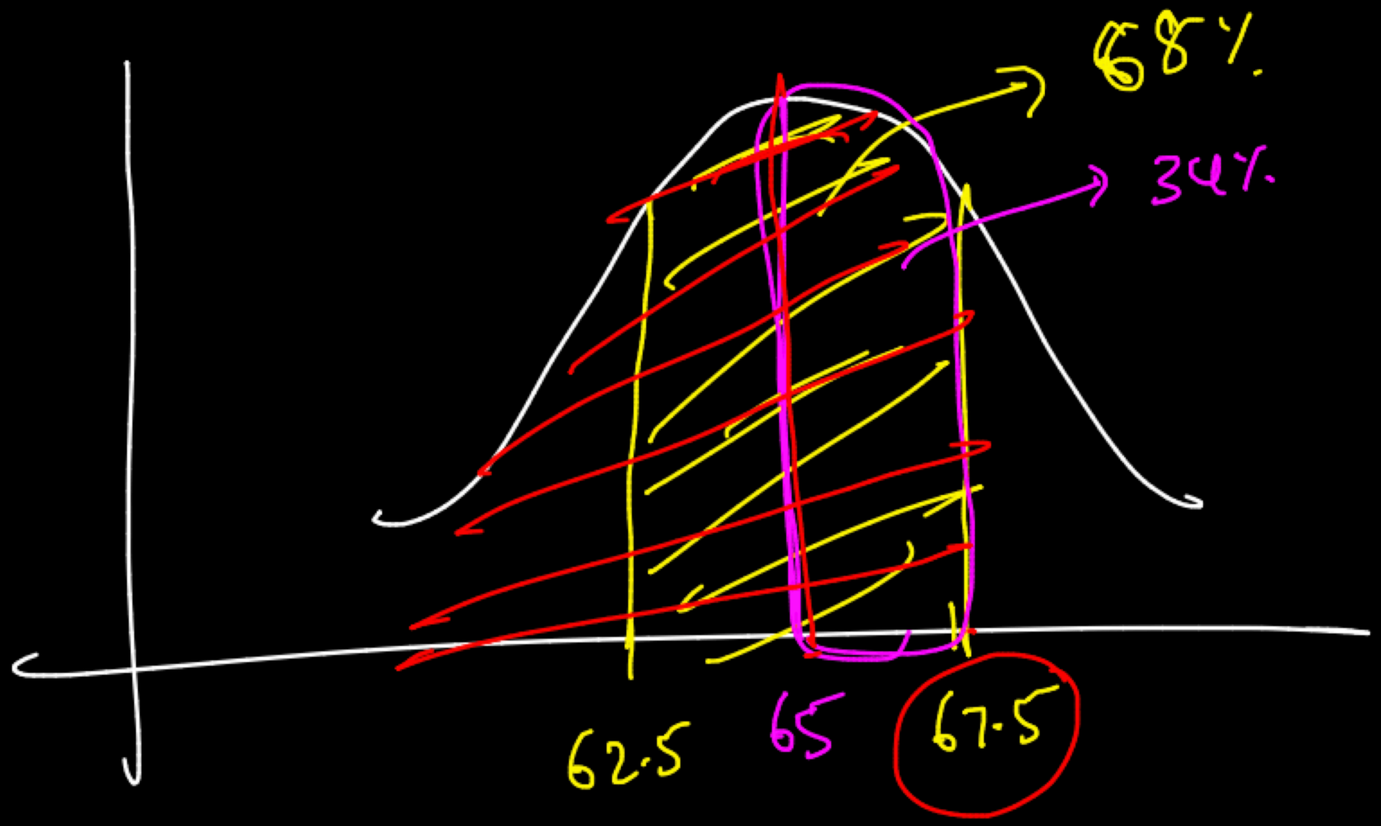
& b/w 65 inches → 67.5 inc

= 0.34

0.5 + 0.34

= 0.

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 7 Jul 10:15 PM





$0.5 + 0.34 = 0.84$

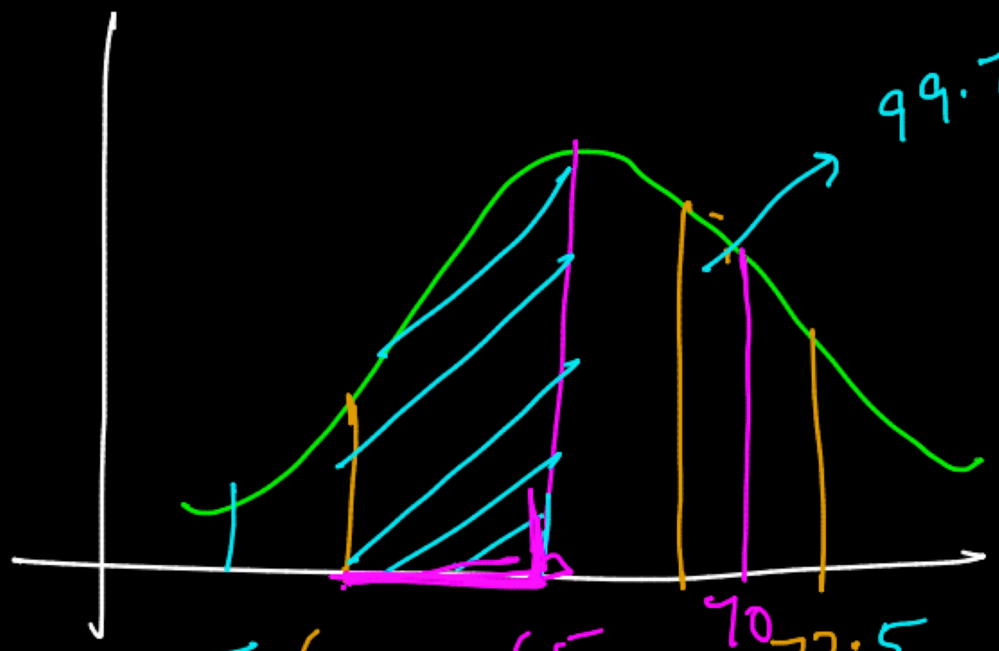
Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 10:29 PM

Empirical

$\Rightarrow \mu = 65$

$\Rightarrow \sigma = 2.5$

Fraction of people whose height is b/w 60 and 72.5?



$99.7/2 = 49.85$

$\frac{95}{2} = 47.5$

16

67.5 — 72.5 $\Rightarrow$ 68%

60 — 70 $\Rightarrow$ 95%

57.5 — 72.5 $\Rightarrow$ 99.7%

$0.475 + 0.4985$

$= 0.9735$

13 / 13

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 7 Jul 10:34 PM

$\mu = 65$

$\sigma = 2.5$

67.5 inch $\Rightarrow$ 16

10 inch

2σ

65 | 67.5 | 70

69.1

67.5

69.1 $\Rightarrow$ How many std dev away from mean

Z-Score

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 10:37 PM

69.1 → "z" std deve away from the mean

$$69.1 = 65 + z * 2.5$$

$$67.5 = 65 + 1 * 2.5$$

$$70 = 65 + 2 * 2.5$$

$$\frac{69.1 - 65}{2.5} = z = \frac{4.1}{2.5} = 1.64$$

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 10:40 PM

69.1 inches ⇒ what fraction of people should than 69.1 inches.

0 → 1

(1) z-table [statisticians]

(2) Python (scipy)

16 / 16

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 7 Jul 10:46 PM

$\mu = 65$

$\sigma = 2.5$

68.5 inches

1.64

$$68.5 = 65 + Z * 2.5$$

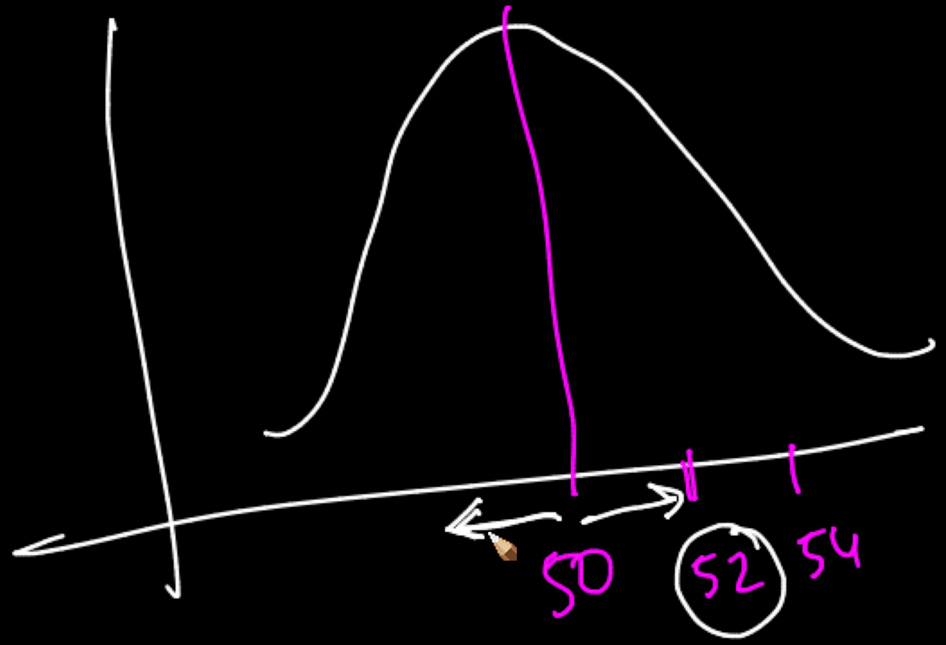
$$Z = \frac{3.5}{2.5} = 1.4$$

17 / 17

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 7 Jul 11:02 PM

$\Rightarrow \mu = 50\ mm$

$\sigma = 2\ mm$



what fraction of balls are smaller than 53 mm.

$53 = 50 + z * 2$

$z = \frac{3}{2} = 1.5$

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 11:06 PM

Derive the Empeical Rule

$\mu = 65$

$\sigma = 2.5$

68

(1) what fraction of people lies b/w 62.5 → 67.5 incl

(i) what fraction of people shorter than 67.5 incl

$67.5 = 65 + z * 2.5$

norm. cdf (1)

$z = 1$

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 10:58 PM

(ii) What fraction of people shorter than 69.5 inch

$$69.5 = 65 + z * 2.5$$

$$-2.5 \Rightarrow z = -1$$

2.5

20 / 20

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 7 Jul 11:06 PM

$\angle = 67.5$

$\angle = 22.5$



21 / 21

Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 7 Jul 11:19 PM

The height of people is Gaussian

mean = 65 inches

σ = 2.5 inches

One person → 96 % of people are shorter than him.

What is his height ?

cdf

PPF

$x = 65 + 1.76 * 2.5$

$= 69.4$